

THERMAL PHYSICS MCQ

1. Thermal physics primarily studies

- a) mechanical energy transfer
- b) temperature and heat
- c) electrical conductivity
- d) magnetic fields

Correct answer: b

2. Temperature measures average

- a) potential energy of particles
- b) kinetic energy of particles
- c) total mass of system
- d) volume change

Correct answer: b

3. Absolute zero equals

- a) 0°C
- b) -273.15°C
- c) 273 K
- d) 100 K

Correct answer: b

4. Kelvin scale relation to Celsius

- a) $\text{K} = ^{\circ}\text{C} + 273.15$
- b) $\text{K} = ^{\circ}\text{C} - 273.15$
- c) $\text{K} = ^{\circ}\text{C} \times 273$
- d) $\text{K} = ^{\circ}\text{C} / 1.8$

Correct answer: a

5. Thermal equilibrium exists when

- a) temperatures differ slightly
- b) no net heat flow occurs
- c) one body hotter than other
- d) volumes equalize

Correct answer: b

6. Heat flows naturally from

- a) cold to hot
- b) hot to cold
- c) equal temperatures
- d) high pressure areas

Correct answer: b

7. SI unit of heat energy

- a) calorie
- b) joule
- c) watt
- d) kelvin

Correct answer: b

8. One calorie raises 1g water by

- a) 1 K
- b) 1°C
- c) 10°C
- d) 100°C

Correct answer: b

9. Heat gained by cold body equals

- a) half heat lost by hot body

- b) heat lost by hot body
- c) double heat lost
- d) total mass change

Correct answer: b

10. Effect of heat includes

- a) mass decrease only
- b) temperature rise and expansion
- c) contraction always
- d) density increase

Correct answer: b

11. Thermal expansion occurs due to

- a) decreased kinetic energy
- b) increased molecular vibration
- c) reduced intermolecular forces
- d) phase change only

Correct answer: b

12. Linear expansion affects

- a) volume only
- b) length only
- c) area only
- d) all dimensions equally

Correct answer: b

13. Coefficient of linear expansion unit

- a) J/K
- b) m/K
- c) K^{-1}

d) $^{\circ}\text{C}/\text{m}$

Correct answer: c

14. $\Delta L = L_0 \alpha \Delta T$ rearranges to α equals

a) $\Delta L / (L_0 \Delta T)$

b) $L_0 / \Delta L$

c) $\Delta T / L_0$

d) $L_0 \Delta T / \Delta L$

Correct answer: a

15. Superficial expansion involves change in

a) length

b) area

c) volume

d) density

Correct answer: b

16. Coefficient of superficial expansion relates to linear as

a) α

b) 2α

c) 3α

d) α^2

Correct answer: b

17. Cubical expansion coefficient for isotropic solids equals

a) α

b) 2α

c) 3α

d) $\alpha/3$

Correct answer: c

18. Liquids show higher expansion than solids because

- a) stronger forces
- b) weaker intermolecular forces
- c) fixed shape
- d) higher density

Correct answer: b

19. Real expansion of liquid excludes

- a) container expansion
- b) liquid volume change
- c) temperature rise
- d) pressure effect

Correct answer: a

20. Apparent expansion appears

- a) larger than real
- b) smaller than real
- c) equal to real
- d) independent of container

Correct answer: b

21. In expansion experiment, liquid level first

- a) rises immediately
- b) falls then rises
- c) remains same
- d) overflows

Correct answer: b

22. Gases expand most because

- a) strong forces between particles

b) large intermolecular distances

c) high density

d) fixed volume

Correct answer: b

23. Boyle's law states at constant temperature

a) $P \propto V$

b) $P \propto 1/V$

c) $V \propto T$

d) $P \propto T$

Correct answer: b

24. Graphical verification of Boyle's law shows

a) straight line

b) hyperbola

c) parabola

d) circle

Correct answer: b

25. Charles's law applies when

a) pressure varies

b) pressure constant

c) volume constant

d) mass changes

Correct answer: b

26. Charles's law graph V vs T gives

a) hyperbola

b) straight line

c) curve

d) horizontal line

Correct answer: b

27. Avogadro's law relates volume to

a) pressure

b) temperature

c) number of molecules

d) density

Correct answer: c

28. Ideal gas assumes

a) intermolecular forces exist

b) zero volume molecules

c) large molecular size

d) attraction between molecules

Correct answer: b

29. Real gases behave ideally at

a) high pressure low temperature

b) low pressure high temperature

c) room conditions only

d) high density

Correct answer: b

30. Ideal gas equation $PV =$

a) RT

b) nRT

c) μRT

d) kT

Correct answer: b

31. Universal gas constant R value

- a) $0.0821 \text{ L atm mol}^{-1} \text{ K}^{-1}$
- b) $8.314 \text{ J mol}^{-1} \text{ K}^{-1}$
- c) both a and b
- d) 273 K

Correct answer: c

32. Boltzmann constant relates to R as

- a) $R = N_A k_B$
- b) $R = k_B / N_A$
- c) $k_B = R N_A$
- d) both a and c

Correct answer: d

33. Zeroth law defines

- a) heat flow
- b) temperature
- c) work
- d) energy

Correct answer: b

34. Thermometer measures

- a) total kinetic energy
- b) average kinetic energy
- c) potential energy
- d) heat capacity

Correct answer: b

35. Fahrenheit to Celsius conversion

- a) $^{\circ}\text{C} = (F - 32) \times 5/9$

b) $^{\circ}\text{C} = \text{F} \times 9/5$

c) $\text{K} = \text{F} + 273$

d) $\text{F} = \text{K} - 459.67$

Correct answer: a

36. At thermal equilibrium, two bodies have

a) different temperatures

b) same temperature

c) same volume

d) same pressure

Correct answer: b

37. Heat capacity depends on

a) mass only

b) temperature change only

c) mass and specific heat

d) volume only

Correct answer: c

38. Kilocalorie equals

a) 1000 calories

b) 100 calories

c) 10 calories

d) 0.001 calorie

Correct answer: a

39. On heating, substances may undergo

a) contraction

b) phase change

c) density increase

d) mass loss

Correct answer: b

40. Coefficient of cubical expansion highest for

a) solids

b) liquids

c) gases

d) all same

Correct answer: c

41. For steel, $\alpha \approx 12 \times 10^{-6} \text{ K}^{-1}$, then γ equals

a) $12 \times 10^{-6} \text{ K}^{-1}$

b) $24 \times 10^{-6} \text{ K}^{-1}$

c) $36 \times 10^{-6} \text{ K}^{-1}$

d) $48 \times 10^{-6} \text{ K}^{-1}$

Correct answer: c

42. Liquid apparent expansion coefficient

a) greater than real

b) less than real

c) equal to real

d) zero

Correct answer: b

43. Boyle's law fails for

a) ideal gases

b) real gases at high pressure

c) low pressure

d) high temperature

Correct answer: b

44. Charles's law constant is

- a) pressure
- b) volume
- c) mass
- d) density

Correct answer: a

45. Avogadro's number value

- a) $6.023 \times 10^{23} \text{ mol}^{-1}$
- b) $6.023 \times 10^{33} \text{ mol}^{-1}$
- c) 6023 mol^{-1}
- d) 6.023 mol^{-1}

Correct answer: a

46. Ideal gas at STP occupies

- a) 11.2 L per mole
- b) 22.4 L per mole
- c) 44.8 L per mole
- d) 1 L per mole

Correct answer: b

47. Real gases deviate from ideal due to

- a) zero volume
- b) intermolecular forces
- c) constant pressure
- d) fixed temperature

Correct answer: b

48. PV/nT equals for ideal gas

- a) zero

- b) R
- c) k_B
- d) 1

Correct answer: b

49. Thermal expansion minimum for

- a) aluminium
- b) invar
- c) copper
- d) brass

Correct answer: b

50. In Boyle's law, if V doubles, P becomes

- a) double
- b) half
- c) same
- d) quadruple

Correct answer: b

51. Charles's law: if T doubles (K), V becomes

- a) half
- b) double
- c) same
- d) quadruple

Correct answer: b

52. Absolute scale avoids

- a) negative temperatures
- b) positive temperatures
- c) zero temperature

d) high temperatures

Correct answer: a

53. Heat transfer modes include

a) conduction only

b) convection only

c) radiation only

d) all three

Correct answer: d

54. Zeroth law leads to

a) first law

b) temperature concept

c) second law

d) third law

Correct answer: b

55. 0°C equals

a) 273.15 K

b) 0 K

c) 32 K

d) 100 K

Correct answer: a

56. Calorie defined for water

a) 1g by 1°C

b) 1 kg by 1°C

c) 1L by 10°C

d) 100g by 1°C

Correct answer: a

57. Superficial expansion $\Delta A =$

- a) $A_0 \beta \Delta T$
- b) $A_0 \alpha \Delta T$
- c) $V_0 \gamma \Delta T$
- d) $L_0 \alpha \Delta T$

Correct answer: a

58. For liquids, real γ equals

- a) apparent γ
- b) apparent $\gamma +$ container γ
- c) container $\gamma -$ apparent γ
- d) zero

Correct answer: b

59. Gas expansion largest coefficient because

- a) molecules far apart
- b) strong bonds
- c) high density
- d) fixed shape

Correct answer: a

60. Ideal gas equation solves for unknown

- a) density
- b) viscosity
- c) any P, V, T, n
- d) molecular weight

Correct answer: c

61. $R = 8.314 \text{ J mol}^{-1} \text{ K}^{-1}$ used in

- a) $PV = nRT$

- b) $PV = RT$
- c) $P = nRT/V$
- d) all gas equations

Correct answer: d

62. Compressibility highest for

- a) solids
- b) liquids
- c) gases
- d) plasmas

Correct answer: c

63. At constant P and T, $V \propto$

- a) $1/n$ (Avogadro)
- b) n (Avogadro)
- c) T (Charles)
- d) $1/P$ (Boyle)

Correct answer: b

64. Plot P vs $1/V$ for Boyle's law gives

- a) curve
- b) straight line
- c) parabola
- d) hyperbola

Correct answer: b

65. Extrapolating Charles's V-T graph intersects

- a) $+273^{\circ}\text{C}$
- b) -273°C
- c) 0°C

d) 100°C

Correct answer: b

66. Invar used in precision instruments because

a) high expansion

b) very low expansion

c) high conductivity

d) low density

Correct answer: b

67. Heat lost = heat gained principle applies to

a) isolated systems

b) closed systems

c) thermal equilibrium

d) calorimetry

Correct answer: d

68. Coefficient values: glass <

a) steel < aluminium

b) aluminium < steel

c) steel = aluminium

d) glass > steel

Correct answer: a

69. Apparent expansion used in

a) thermometers

b) designing liquid containers

c) gas cylinders

d) solid rods

Correct answer: b

70. Real gases liquefy because

- a) ideal behaviour
- b) intermolecular attractions
- c) zero volume
- d) high temperature

Correct answer: b

71. PV/T constant for

- a) Boyle
- b) Charles
- c) combined gas law
- d) Avogadro

Correct answer: c

72. 1 atm =

- a) 760 mm Hg
- b) 103 kPa
- c) both
- d) 1 Pa

Correct answer: c

73. STP conditions

- a) 0°C, 1 atm
- b) 25°C, 1 atm
- c) 0 K, 1 Pa
- d) 273 K, 760 torr

Correct answer: a

74. Molar volume ideal gas STP

- a) 22.4 L

b) 11.2 L

c) 44.8 L

d) 2.24 L

Correct answer: a

75. k_B value

a) 1.38×10^{-23} J/K

b) 8.314 J/mol K

c) 6.023×10^{23}

d) 0.0821 L atm

Correct answer: a

76. $N_A k_B =$

a) R

b) PV

c) T

d) n

Correct answer: a

77. Thermal expansion anisotropic materials

a) same all directions

b) different directions

c) zero

d) infinite

Correct answer: b

78. Liquid thermometer uses

a) mercury expansion

b) alcohol contraction

c) gas compression

d) solid linear expansion

Correct answer: a

79. Boyle's law isothermal means

a) T constant

b) P constant

c) V constant

d) n constant

Correct answer: a

80. Failure of gas laws at high P due to

a) molecular volume negligible

b) attractions negligible

c) molecular volume significant

d) ideal behaviour

Correct answer: c